



II. PROPOSED SYSTEM 1

1. Proposed System Title

Customo: A Customizable Ordering System for Food Establishments

2. System Overview

Briefly describe what the proposed system is and what it is intended to do.

Customo is a self-service, event-driven ordering system built for fast food chains, casual dining spots, and restaurants that still rely on cardboard menus or verbal, staff-relayed orders. Instead of telling a server their allergies or requests by word of mouth, customers use a tablet or kiosk-style interface to browse the menu, tag ingredients they are allergic to, and pick add-ons or extras for each item they order.

Every action the customer takes, such as selecting a dish, flagging an allergy, or adding an extra, triggers an event that updates the order summary and, once submitted, notifies the kitchen directly. This removes the guesswork and miscommunication that happens when allergy or add-on requests are passed along by memory from customer to server to kitchen staff.

The goal is simple: fewer wrong orders, faster service, and a more convenient, safer experience for both the customer and the restaurant staff.

3. Problem Statement

What problem does the system aim to solve?

Many restaurants in the Philippines, especially those still using cardboard menus or manual, verbal ordering, run into repeated complaints from customers about allergic reactions or wrong orders because their allergy or dietary requests were not properly communicated to the kitchen.

Add-on requests also slow down service, since customers usually have to flag down a server, wait for them to be free, then wait again for the request to be relayed and confirmed. This creates delays and a less convenient overall dining experience.

Customo aims to solve both problems by letting customers directly and accurately input their allergy concerns and add-on preferences into the system itself, cutting out the risk of miscommunication and the wait time that comes with manual relay.

4. Target Users

Small to mid-sized fast food chains, casual dining spots, and fine dining restaurants in the Philippines that still use manual or cardboard-based ordering; and their dine-in customers, particularly those with food allergies or specific dietary needs.

5. Proposed Features

- **Allergy/ingredient tagging:** customers can flag ingredients they are allergic to per dish before ordering.
- **Add-on and customization selector:** pick extras, size, or ingredient swaps for any order item.
- **Real-time order summary:** shows the finalized customized order before submission, so customers can double-check it.
- **Automatic kitchen notification:** allergy tags and add-ons are sent straight to the kitchen display once the order is confirmed, triggered as an event.
- **Event-based price recalculation:** the total updates instantly whenever an add-on is selected or removed.
- **Table selection:** customers choose their table number before ordering, so the kitchen knows where each order belongs.
- **Allergy warning on dishes:** dishes that contain an allergen the customer declared are highlighted, and a warning appears before they can be added to the order.

6. Scope and Limitations

Scope:

The system covers the customer-facing ordering process for a single restaurant branch (dine-in): browsing the menu, tagging allergies, choosing add-ons, reviewing the order summary, and submitting the order, which then triggers a kitchen notification. It also includes a kitchen display that receives submitted orders in real time. Platform: Web-based, designed for tablet/kiosk use at the table or counter.

Limitations:

Customo does not handle payment processing, inventory tracking, or syncing across multiple branches in this version. It also depends on kitchen staff manually reading and acting on the allergy flags shown on their display; it does not automatically cross-check allergens against a restaurant's actual ingredient stock. In this version, menu data is preloaded into the database and there is no administrator interface for editing it.

Possible Future Enhancements:

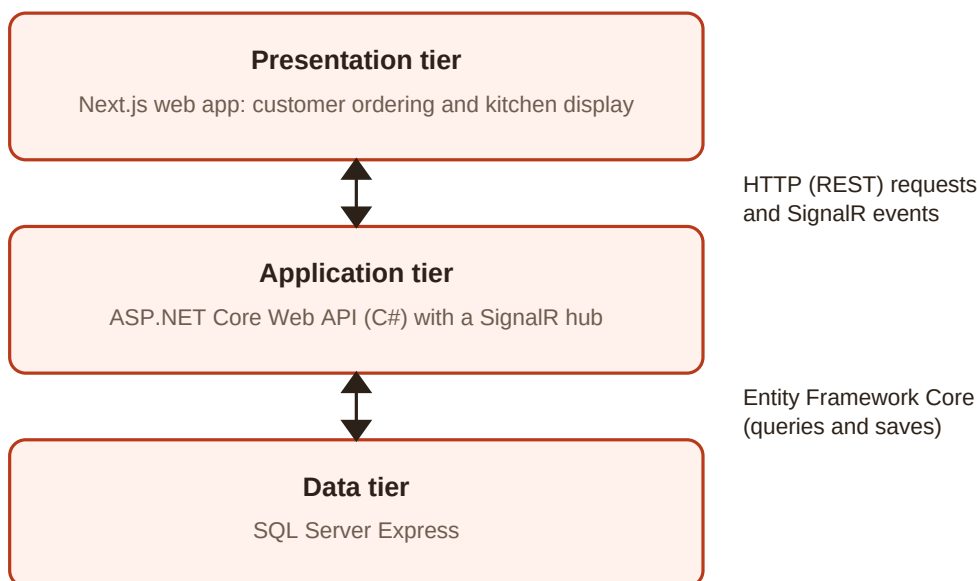
- **Payment processing:** integrate a payment gateway (cash, card, or e-wallets like GCash) directly into Customo instead of handling this separately at the counter.
- **Inventory tracking:** link the system to a live inventory database so out-of-stock items or ingredients auto-disable on the menu.

- **Multi-branch syncing:** move menu, pricing, and allergy data to a shared cloud database so all branches stay consistent and updates only need to be made once.
- **Allergy flag safeguards:** highlight allergy-flagged orders in a distinct color or with an alert on the kitchen display, and require staff to tap "acknowledge" before the order is marked as accepted, reducing the chance of a flag being missed.
- **Admin menu management:** let authorized staff add, edit, or remove menu items, prices, add-ons, and allergen tags without changing the database directly.

7. Technical Design

7.1 System Architecture

Customo follows a three-tier client-server architecture with an event-driven layer for real-time updates. The presentation tier is a web application used by customers (ordering) and kitchen staff (order display). The application tier is a web API that validates and processes orders. The data tier is a relational database that stores the menu and all submitted orders. The web application and the API are separate programs that communicate only through HTTP requests and SignalR messages.



7.2 Technology Stack

- **Frontend:** Next.js (React) with TypeScript and Tailwind CSS
- **Backend:** ASP.NET Core Web API (C#, .NET)
- **Real-time communication:** SignalR
- **Data access:** Entity Framework Core (code-first migrations)
- **Database:** Microsoft SQL Server Express
- **Tools:** Visual Studio / VS Code, .NET CLI, npm, SQL Server Management Studio

7.3 Database Design

The database has nine tables. Menu data is stored in **Categories**, **MenuItems**, **Allergens**, **MenuItemAllergens** (which dish contains which allergen), and **MenuItemOptions** (sizes, extras, and swaps with their price changes). Order data is stored in **Orders**, **OrderAllergies** (the allergies the customer declared), **OrderItems**, and **OrderItemOptions**. The menu is preloaded through seed data.

7.4 Order Processing Flow

1. The customer selects a table and declares allergies, then browses the menu. Dishes that contain a declared allergen are highlighted and show a warning before they can be added.
2. The customer customizes each dish (size, extras, swaps). The order total updates instantly with each change.
3. On the review screen, the customer checks the order and confirms it.
4. The API validates the request, recalculates all prices from the database, marks items that conflict with the customer's allergies, and saves the order.
5. The API then broadcasts an OrderSubmitted event through SignalR. Every open kitchen screen displays the new order immediately, with the allergy flags highlighted.

7.5 Design Decisions

- Prices are always calculated by the server, so a modified client cannot change the total.
- Each order stores the price and option names at the time of ordering, so later menu changes do not alter past orders.
- Allergy conflicts are checked twice: by the interface to warn the customer, and by the server to flag the order for the kitchen.
- The system is split into a separate frontend and backend so that either can be changed or deployed independently.